3 Counting and Probability

The big idea. In this topic every problem comes back to *one core skill*: **counting correctly** – missing nothing and counting nothing twice.

- **Counting** answers the question "*how many?*": how many numbers, how many seating arrangements, how many rectangles in a figure, how many toothpicks...
- **Probability** answers the question "*how likely is it?*". When all outcomes are equally likely, we only need to *count twice*:

$$\text{Probability} = \frac{\text{number of favourable outcomes}}{\text{number of all possible outcomes}}.$$

- Our counting tools: **systematic listing**, **tables**, **tree diagrams**, the **multiplication rule**, **splitting into cases** (the addition rule), **finding a pattern** and **reasoning by elimination**.

Always ask yourself three questions before starting: *(1) How many possible outcomes are there altogether? (2) Are these outcomes "fair" (equally likely)? (3) How many of them satisfy the requirement?*



① Experiments, outcomes, events and probability

Key words.

- A **random experiment** is an action for which we know in advance *all the possible outcomes*, but not *which outcome will happen*. Examples: rolling a die, tossing a coin, spinning a spinner, drawing a marble from a bag without looking.
- The **set of possible outcomes** is the list of *all* outcomes. Rolling a die: $\{1, 2, 3, 4, 5, 6\}$ – 6 outcomes. Tossing a coin: {heads, tails} – 2 outcomes.
- An **event** is something we are interested in, for example "rolling an even number". The outcomes that make the event happen are called **favourable outcomes** for the event. The event "rolling an even number" has 3 favourable outcomes: $2, 4, 6$.
- Outcomes are **equally likely** if none of them is "easier to get" than another (a fair die, a fair coin, equal sectors on a spinner, marbles of identical shape and size).

Probability formula (when the outcomes are equally likely):

$$P(\text{event}) = \frac{\text{number of outcomes favourable to the event}}{\text{number of all possible outcomes}}.$$

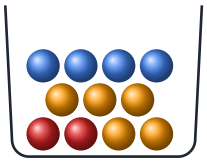
Properties. The number of favourable outcomes is between 0 and the total number of outcomes, so always $0 \le P \le 1$.

- $P = 0$: an *impossible* event (for example rolling a 7 on a die).
- $P = 1$: a *certain* event (for example rolling a number less than 7).

Why is it true? Why a division? Suppose there are n equally likely outcomes. The total "chance" 1 (one of the outcomes is sure to happen) is shared *equally* among n outcomes, so each outcome has probability $\frac{1}{n}$. If the event has k favourable outcomes, we add k equal parts: $\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n} = \frac{k}{n}$. □



Spinner with 6 equal sectors: each sector has probability $\frac{1}{6}$



Bag with 2 red, 5 yellow, 4 blue: $P(\text{red}) = \frac{2}{11}$

Common mistake. Count *objects*, not *kinds*. The bag above has 3 *colours* but 11 *marbles*. The probability of drawing a red marble is $\frac{2}{11}$, **not** $\frac{1}{3}$. Each marble is one equally likely outcome; the three colours are not equally likely (yellow has more marbles, so it is easier to draw).

Common mistake. Use the formula only when the outcomes are equally likely. On a spinner with big and small sectors, the sectors are *not* equally likely. Then the probability of landing in a sector equals *the fraction of the whole circle* that the sector takes up, that is $\frac{\text{central angle of the sector}}{360°}$. For example a sector with angle 90° has probability $\frac{90}{360} = \frac{1}{4}$.

ThinkAlgo Tip. In multiple-choice questions the answer may be written in *unsimplified* form, such as $\frac{2}{6}$ instead of $\frac{1}{3}$. Compare the values of the fractions, not just how they look. On the other hand, when you write your own solution, simplify your final answer.

② Complementary events. Finding a quantity from a probability

Complementary event. The event "A does *not* happen" is called the complement of A. We have

$$P(\text{not } A) = 1 - P(A).$$

Example: rolling a die, $P(\text{a } 6) = \frac{1}{6}$, so $P(\text{not a } 6) = 1 - \frac{1}{6} = \frac{5}{6}$.

Finding a quantity from a probability. From the probability formula we get

$$\text{number of favourable outcomes} = P \times \text{number of all outcomes}.$$

Example: a bag has 20 sweets and the probability of drawing a lemon sweet is $\frac{3}{4}$; then there are $20 \times \frac{3}{4} = 15$ lemon sweets.

When the quantities change (drawn and *not replaced*, eaten, added…): **recount both the numerator and the denominator** before computing the probability for the next draw.

Why is it true? Suppose there are n outcomes, of which k are favourable to A. Each of the remaining $n - k$ outcomes makes A *not* happen. So $P(\text{not } A) = \frac{n-k}{n} = \frac{n}{n} - \frac{k}{n} = 1 - P(A)$. □

Common mistake. A box has 10 red and 10 blue marbles. A student takes out 1 blue marble and does *not put it back*. What is the probability that the next marble is red?

- **Wrong:** "still $\frac{10}{20}$" (forgetting that the box now has one marble fewer).
- **Right:** 19 marbles are left, and all 10 red ones are still there, so the probability is $\frac{10}{19}$.

Note for parents. Do a real experiment with sweets or bottle caps of different colours: put them in a cloth bag, count together "how many are there altogether, how many are red", then take one out and ask the same two questions again. Your child will see for themselves that both the numerator and the denominator change.

③ Systematic listing: lists, tables and tree diagrams

When the number of possibilities is not too large, the safest method is to **list them all**. To list without missing or repeating anything, we need **an order**:

- An **ordered list** (from smallest to largest, alphabetically…). Example: the two-digit numbers with different digits made from 1, 2, 3: 12, 13, 21, 23, 31, 32 – ordered by the tens digit.
- A **two-way table** when the experiment has *two stages* (rolling two dice, choosing a main dish and a dessert…): each row is an outcome of stage 1, each column an outcome of stage 2, each cell a combined outcome.
- A **tree diagram** when there are *two, three, four stages in a row*: each stage branches out; each "path" from the root to the end of a branch is one outcome.

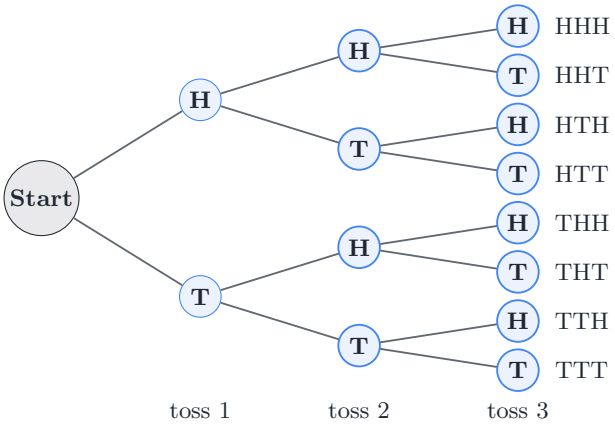
Table for rolling two dice (one red, one blue). Each cell shows the *sum* of the two numbers. There are 6 rows and 6 columns, so $6 \times 6 = 36$ equally likely outcomes.

red ↓ \ blue →	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Orange cells: sum equals 7. There are 6 such cells, so $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$.

Common mistake. (1, 2) and (2, 1) are two different outcomes! Red 1, blue 2 is different from red 2, blue 1. If you only count "pairs of numbers" without order, the pairs are *no longer* equally likely (the pair $\{1, 2\}$ can happen in 2 ways, the pair $\{1, 1\}$ in only 1 way). That is why the sums $2, 3, \ldots, 12$ are *not* equally likely: a sum of 7 can happen in 6 ways, a sum of 2 in only 1 way.

Tree diagram for tossing a coin three times (H: heads, T: tails). Each toss has 2 branches, so there are $2 \times 2 \times 2 = 8$ paths – that is, 8 equally likely outcomes.



Example: the event "exactly two heads" consists of HHT, HTH, THH – 3 outcomes, so its probability is $\frac{3}{8}$.

ThinkAlgo Tip. A problem like "walk on a grid by tossing a coin" (heads: move up, tails: move right) is really a problem about counting sequences of heads and tails: after 4 tosses there are $2^4 = 16$ equally likely sequences, and the *end point* depends only on the *number of heads*, not on their order.

④ The multiplication rule and the addition rule

Multiplication rule. If a task consists of two successive stages, stage 1 can be done in m ways, and *for each* way of doing stage 1, stage 2 can be done in n ways, then the whole task can be done in $m \times n$ ways. (With three or four stages: keep multiplying.)

Addition rule (splitting into cases). If the ways are split into cases that *do not overlap*: case 1 has a ways, case 2 has b ways, ..., then the total number of ways is $a + b + \ldots$

Common situations with digits.

- Three-digit numbers made from 3 given non-zero digits, *no repetition*: $3 \times 2 \times 1 = 6$ numbers.
- Three-digit numbers made from 3 given non-zero digits, *repetition allowed*: $3 \times 3 \times 3 = 27$ numbers.
- The first (leading) digit **cannot be** 0: 9 choices (1 to 9); the other places have 10 choices (0 to 9).
- A three-digit **palindrome** (reads the same forwards and backwards) has the form $\overline{aba}$: 9 ways to choose a, 10 ways to choose b, and the last digit is "forced" to equal a (only 1 way). There are $9 \times 10 = 90$ such numbers.



Number of three-digit palindromes: $9 \times 10 \times 1 = 90$.

Why is it true? Why is the multiplication rule true? Draw a table with m rows (the ways of doing stage 1) and n columns (the ways of doing stage 2). Each cell of the table is *exactly one* way of doing the whole task, and each way of doing the whole task is in *exactly one* cell. The table has m rows of n cells, so $m \times n$ cells. A tree diagram gives the same result: m big branches, each splitting into n small branches. □

Choosing two objects (order does not matter). Choosing 2 of n students for classroom duty: if we care about order (first person, second person) there are $n \times (n-1)$ ways; but each pair is counted *twice* (An–Bình and Bình–An), so the number of pairs is

$$\frac{n \times (n-1)}{2}.$$

Example: there are $\dfrac{10 \times 9}{2} = 45$ pairs of different numbers taken from 10 numbers.

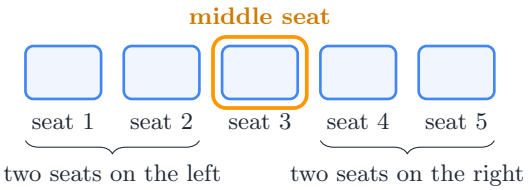
Common mistake. Double counting. When the question asks "how many *pairs*", "how many *groups*" (who comes first does not matter) and you use the multiplication rule with order, each group is counted several times – you must divide. On the other hand, when the question is about *numbers* (the order of the digits matters) or *seating in a row*, order matters and you do *not* divide.

ThinkAlgo Tip. When splitting into cases, ask: "Do the cases *cover* every possibility? Do they *overlap*?" For example, for the sum $m + n$ to be even, m and n must be *both odd* or *both even* – these two cases are separate and together they cover every possibility.

⑤ Reasoning by elimination: seating and colouring

Some problems do not ask "how many" but "*who can, who cannot*", "*what is the least number needed*". Then we use **reasoning**:

- **Draw a diagram** of the positions (seats, cells, faces…) and try filling it in.
- Two people who **must sit together** can be treated as **one block** taking two neighbouring seats.
- To show "*X can* sit in the middle seat": just **give one arrangement** that satisfies all the conditions.
- To show "*X cannot* sit in the middle seat": **suppose** X sits in the middle seat and show that *every* remaining arrangement breaks a condition (leading to a contradiction). This is called **proof by contradiction**.
- A "what is the *least* number" problem has **two parts**: (1) show a way that works with that number; (2) explain why fewer is impossible.



Key observation: once someone sits in the middle seat, the other four seats split into *two pairs of neighbouring seats* (seats 1–2 and seats 4–5). The two seats on either side of the middle seat (seat 2 and seat 4) are *not* next to each other.

Colouring so that neighbouring parts have different colours. Two faces of a cube are "neighbours" if they share an edge. Each face is a neighbour of 4 faces and is *not* a neighbour of exactly 1 face – the **opposite** face. A cube has 3 pairs of opposite faces (top–bottom, front–back, left–right).

- 3 *colours are enough:* paint each pair of opposite faces one colour; two faces of the same colour are opposite, so they are not neighbours.
- *Fewer than* 3 *colours is impossible:* the three faces meeting at a corner are neighbours of each other in pairs, so they need three different colours.

ThinkAlgo Tip. For a multiple-choice question "Who *cannot*…?", you can work backwards: for each option, try to find *one* valid arrangement. Every option for which you find an arrangement is eliminated; the remaining option is the answer – but remember to explain why it really cannot be arranged.

⑥ Counting shapes in a figure

Principle. Counting shapes is a listing problem. To miss nothing and count nothing twice, **split into types** (by size, by position) and then **add up**.

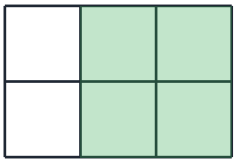
The "choose two lines" trick. A rectangle in a grid is completely determined by *two horizontal*

lines (top and bottom sides) and *two vertical lines* (left and right sides). Therefore:

- A strip of 1 row and n squares has $n+1$ vertical lines; the number of rectangles is the number of ways to choose 2 vertical lines: $\frac{(n+1)\times n}{2}$. Example: a strip of 4 squares has $\frac{5\times 4}{2} = 10$ rectangles.
- A grid with a horizontal lines and b vertical lines (all running the full length) has $\frac{a(a-1)}{2} \times \frac{b(b-1)}{2}$ rectangles (multiplication rule!).
- The same counting works for *parallelograms* in a grid whose slanted lines are parallel to each other.



Choose vertical lines 2 and 4
$\Rightarrow$ the orange rectangle

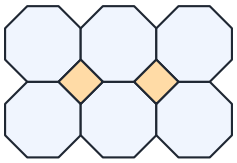


3 horizontal, 4 vertical lines:
$3 \times 6 = 18$ rectangles

Why is it true? Why does a strip of n squares have $\frac{(n+1)\times n}{2}$ rectangles? Choose the left side: $n+1$ ways; choose the right side (different from the left side): n ways – that gives $(n+1)\times n$ *ordered* choices. But choosing "line 2 then line 4" or "line 4 then line 2" gives the same rectangle, so each rectangle is counted 2 times; divide by 2. Counting by size gives the same result: a strip of 4 squares has 4 rectangles of width 1, 3 of width 2, 2 of width 3 and 1 of width 4, in total $4+3+2+1 = 10$. □

Common mistake. When the shapes **partly overlap** (not a grid with full-length lines), the "choose two lines" trick *does not work*, because many segments do not run all the way across. Check each shape: **are all four sides really drawn?** Do not count "imaginary rectangles" that are missing a side.

Counting in a tiling. A tiling made of regular octagons and small squares: if there are m rows with n octagons in each row, there are $m \times n$ octagons. Each small square sits where *four octagons meet*; between n columns there are $n-1$ gaps and between m rows there are $m-1$ gaps, so there are $(m-1)\times(n-1)$ squares.



2 rows, 3 columns: $2 \times 3 = 6$ octagons,
$(2-1)\times(3-1) = 2$ squares.

⑦ Counting with patterns: evenly spaced sequences, constant rates

Counting the whole numbers from a to b (including a and b): there are $b - a + 1$ numbers. For example from 1 to 11 there are 11 numbers, of which 6 are odd $(1, 3, 5, 7, 9, 11)$ and 5 are even.

Evenly spaced sequence (each number is a fixed amount d more than the one before):

$$\text{number of terms} = \frac{\text{last} - \text{first}}{d} + 1, \qquad \text{sum} = \frac{(\text{first} + \text{last}) \times \text{number of terms}}{2}.$$

Quantities that change at a constant rate (moving steadily, flowing steadily, counting steadily…): amount = rate × time. For example, if 3 sheep pass each minute, then in 14 minutes $3 \times 14 = 42$ sheep pass.

Figures made of sticks. Count the *horizontal sticks* and the *vertical sticks* separately and then add – usually easier than counting square by square.

Events that repeat in cycles. If one thing repeats every a minutes and another every b minutes, and they start together, they coincide again every $\text{LCM}(a, b)$ minutes.

Why is it true? Gauss's way of adding. Compute $S = 1 + 2 + 3 + \cdots + 10$. Write the sum twice, once forwards and once backwards:

$$\begin{array}{rcccccc} S &=& 1 & +2 & +3 & +\cdots+ & 10 \\ S &=& 10 & +9 & +8 & +\cdots+ & 1 \\ \hline 2S &=& 11 & +11 & +11 & +\cdots+ & 11 \end{array}$$

There are 10 columns, each with sum 11, so $2S = 10 \times 11 = 110$ and $S = 55$. In general: sum $= \dfrac{(\text{first} + \text{last}) \times \text{number of terms}}{2}$. □

Common mistake. The "off by one" error. From 3 to 10 there are $10 - 3 + 1 = 8$ numbers, *not* 7. A fence 10 m long with a post every 1 m (including both ends) needs 11 posts, not 10. Test a formula on a small example to check it.

Note for parents. For "toothpick" problems, let your child actually build the small figures (1 square, 2 squares, 3 squares…) with toothpicks, write the number of toothpicks in a table, and then ask: "each time we add one square, how many toothpicks do we add?". A child who finds the pattern alone remembers it much longer than a memorised formula.

Summary: read the clues, choose the tool.

Clue in the problem	Tool to use
"choose one at random", "what is the probability"	count all outcomes and favourable outcomes, then divide
probability given, quantity asked	favourable $= P \times$ total
something was taken out/eaten before the next draw	recount numerator and denominator
"at least one", "not"	complement: $1 - P$
two dice, two independent choices	two-way table
tossing a coin several times, several stages in a row	tree diagram
"how many numbers can be made", "how many ways"	multiplication rule (choose restricted positions first)
"how many pairs" (order does not matter)	multiplication rule, then divide by 2
conditions of the form "either... or..."	split into cases, add
"who can / cannot"	try one arrangement / proof by contradiction
"how many shapes..."	split by size, or choose two lines
rows of seats, toothpicks, constant rates	find the pattern, evenly spaced sequences

Example 1

A box contains 3 red marbles, 5 blue marbles and 4 yellow marbles, all of the same size. One marble is drawn at random.

a) Find the probability of drawing a blue marble.

b) Find the probability of drawing a marble that is *not* red.

c) Find the probability of drawing a purple marble.

Guided solution.

Step 1. Count all possible outcomes

Each marble is one outcome, and since the marbles have the same size the outcomes are equally likely. Total number of marbles: $3 + 5 + 4 = 12$. So there are 12 possible outcomes.

Step 2. Part a

There are 5 blue marbles, i.e. 5 favourable outcomes. $P(\text{blue}) = \dfrac{5}{12}$.

Step 3. Part b – use the complement

$P(\text{red}) = \dfrac{3}{12} = \dfrac{1}{4}$, so $P(\text{not red}) = 1 - \dfrac{1}{4} = \dfrac{3}{4}$.

Another way (direct counting): the marbles that are not red are 5 blue and 4 yellow, i.e. 9 marbles; $P = \dfrac{9}{12} = \dfrac{3}{4}$. Both ways give the same result.

Step 4. Part c

There are no purple marbles in the box: 0 favourable outcomes. $P(\text{purple}) = \dfrac{0}{12} = 0$ – an impossible event.

Answer: a) $\frac{5}{12}$; **b)** $\frac{3}{4}$; **c)** 0.

Example 2

A bag contains 30 sweets, some strawberry and some orange. If one sweet is drawn at random, the probability that it is strawberry is $\frac{2}{5}$.

a) How many strawberry sweets and how many orange sweets are in the bag?

b) An takes out 2 strawberry sweets and eats them. Then Bình draws one sweet at random from the bag. Find the probability that Bình gets a strawberry sweet.

Guided solution.

Step 1. Find the number of strawberry sweets

Recall. Number of favourable outcomes = probability × number of all outcomes.

Strawberry sweets: $30 \times \frac{2}{5} = \frac{30 \times 2}{5} = \frac{60}{5} = 12$. Orange sweets: $30 - 12 = 18$.

Check: $\frac{12}{30} = \frac{2}{5}$ ✓.

Step 2. Recount after An eats 2 sweets

	strawberry	orange	total
At the start	12	18	30
After An eats 2 strawberry	$12 - 2 = 10$	18	$30 - 2 = 28$

Step 3. Compute the probability for Bình

$P(\text{strawberry}) = \frac{10}{28} = \frac{5}{14}$.

Common mistake. Do not answer $\frac{10}{30}$: An has eaten 2 sweets, so the total number of sweets (the denominator) has gone down too.

Answer: a) 12 strawberry, 18 orange; b) $\frac{5}{14}$.

Example 3

A red die and a blue die (both fair) are rolled together. Find the probability of each event:

a) The sum of the two numbers is 8.

b) Both dice show the same number.

c) The sum of the two numbers is greater than or equal to 10.

Guided solution.

Step 1. Make a table of all outcomes

Rows are the numbers on the red die, columns the numbers on the blue die, and each cell shows the sum. The table has $6 \times 6 = 36$ cells, corresponding to 36 equally likely outcomes.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

sum equals 8 (5 cells)
○ both numbers equal (6 cells)
sum ≥ 10 (6 cells)

Step 2. Part a

The cells with sum 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$ – 5 outcomes (the first number is the red die). $P = \frac{5}{36}$.

Step 3. Part b

Equal numbers lie on the diagonal: $(1,1), (2,2), \ldots, (6,6)$ – 6 outcomes. $P = \frac{6}{36} = \frac{1}{6}$.

Step 4. Part c

Sum 10: $(4,6), (5,5), (6,4)$; sum 11: $(5,6), (6,5)$; sum 12: $(6,6)$. Altogether $3+2+1 = 6$ outcomes. $P = \frac{6}{36} = \frac{1}{6}$.

ThinkAlgo Tip. The 6×6 table also shows *symmetry*: the number of cells above the diagonal (red smaller than blue) equals the number of cells below the diagonal (red greater than blue).

Answer: a) $\frac{5}{36}$; b) $\frac{1}{6}$; c) $\frac{1}{6}$.

Example 4

A fair coin is tossed three times in a row. Find the probability of getting: a) exactly two heads; b) at least one head.

Guided solution.

Step 1. Draw a tree diagram and list

Each toss has 2 outcomes (H or T). By the tree diagram in section 3, there are $2 \times 2 \times 2 = 8$ equally likely outcomes:

$$\text{HHH},\ \text{HHT},\ \text{HTH},\ \text{HTT},\ \text{THH},\ \text{THT},\ \text{TTH},\ \text{TTT}.$$

Step 2. Part a

The outcomes with exactly two H's: HHT, HTH, THH – 3 outcomes. $P = \frac{3}{8}$.

Step 3. Part b – complement

The complement of "at least one head" is "no heads at all", which is only TTT (1 outcome). So $P(\text{at least one H}) = 1 - \frac{1}{8} = \frac{7}{8}$.

ThinkAlgo Tip. When you see the words "*at least one*", think immediately of the complement "*none at all*" – it usually has very few outcomes and is easy to count.

Answer: a) $\frac{3}{8}$; b) $\frac{7}{8}$.

Example 5

Using the digits $1, 2, 3, 4$, how many three-digit numbers can be made if: a) digits may be repeated; b) the digits must be different; c) the digits are different and the number is even?
d) One of the numbers from part b) is chosen at random. Find the probability that it is even.

Guided solution.

Step 1. Part a – repetition allowed
Hundreds: 4 ways; tens: still 4 ways; ones: 4 ways. By the multiplication rule: $4 \times 4 \times 4 = 64$ numbers.

Step 2. Part b – no repetition

$$\underbrace{4}_{\text{hundreds: any}} \times \underbrace{3}_{\text{tens: not the hundreds digit}} \times \underbrace{2}_{\text{ones: not the two above}} = 24$$

Hundreds: 4 ways. The tens digit must differ from the digit already used: 3 ways. The ones digit must differ from both digits used: 2 ways. There are $4 \times 3 \times 2 = 24$ numbers.

Step 3. Part c – even, no repetition: choose the ones digit FIRST
An even number ends in 2 or 4: 2 ways. Then the hundreds digit is one of the 3 remaining digits: 3 ways; the tens digit: 2 ways. There are $2 \times 3 \times 2 = 12$ numbers.

ThinkAlgo Tip. A position with a *condition* (must be even, must not be 0...) should be **chosen first**. If you choose the hundreds digit first, the number of choices for the ones digit is sometimes 1 and sometimes 2 (depending on whether the hundreds digit used an even or an odd digit) – very easy to get wrong.

Step 4. Part d
$P(\text{even}) = \dfrac{12}{24} = \dfrac{1}{2}$.

Check: List the numbers starting with 1: $123, 124, 132, 134, 142, 143$ – exactly $6 = 3 \times 2$ numbers, of which $124, 132, 134, 142$ are even (4 numbers). The four possible first digits behave alike, so there are $4 \times 6 = 24$ numbers, ... matching part b.

Answer: a) 64; b) 24; c) 12; d) $\frac{1}{2}$.

Example 6

Four friends An, Bình, Chi and Dũng sit on four chairs in a row, numbered $1, 2, 3, 4$ from left to right. We know that: Dũng sits on chair 1; Bình and Chi sit next to each other; An does not sit next to Bình; An does not sit on chair 4. Who sits on chair 4?

Guided solution.

Step 1. Place Dũng

Dũng is on chair 1, leaving chairs $2, 3, 4$ for An, Bình and Chi.

Step 2. Consider An's position

An does not sit on chair 4, so An sits on chair 2 or chair 3.

- *If An sits on chair 3:* Bình and Chi take chairs 2 and 4 – these chairs are not next to each other, which breaks the condition "Bình and Chi sit next to each other". Rejected.
- *If An sits on chair 2:* Bình and Chi take chairs 3 and 4 (next to each other, good). An must not be next to Bình, and chair 3 is next to chair 2, so Bình is not on chair 3. So Bình is on chair 4 and Chi on chair 3.

Dũng	An	Chi	Bình
chair 1	chair 2	chair 3	chair 4

Check: Dũng on chair 1 ✓; Bình (chair 4) next to Chi (chair 3) ✓; An (chair 2) is next to Dũng and Chi, not next to Bình ✓; An is not on chair 4 ✓.

Answer: Bình sits on chair 4.

Example 7

a) The figure on the left is a strip of 4 squares. How many rectangles (including squares) are there altogether?
b) The figure on the right is a 2×2 grid of squares. How many rectangles are there? How many of them are squares?



Guided solution.

Step 1. Part a – count by width

Measure the width in squares:

Width	1 square	2 squares	3 squares	4 squares
Number of shapes	4	3	2	1

Total: $4 + 3 + 2 + 1 = 10$ rectangles. *Check with the "choose two vertical lines" trick:* there are 5 vertical lines, and the number of ways to choose 2 of them is $\frac{5 \times 4}{2} = 10$ ✓.

Step 2. Part b – choose two horizontal and two vertical lines

The grid has 3 horizontal and 3 vertical lines. Choose 2 horizontal lines: $\frac{3 \times 2}{2} = 3$ ways; choose 2 vertical lines: 3 ways. By the multiplication rule there are $3 \times 3 = 9$ rectangles.

Step 3. Count the squares separately

1×1 squares: 4; 2×2 squares: 1 (the whole grid). There are $4 + 1 = 5$ squares.

Check: Counting by size: 1×1: 4; 1×2 (lying flat): 2; 2×1 (standing): 2; 2×2: 1. Total $4 + 2 + 2 + 1 = 9$ ✓.

Answer: a) 10 rectangles; b) 9 rectangles, of which 5 are squares.

Example 8

Toothpicks are used to make a row of joined squares as shown below.



1 sq. 2 sq. 3 sq.

a) How many toothpicks are needed for a row of 20 squares?
b) Lan builds, one after another and separately, a row of 1 square, a row of 2 squares, ..., a row of 10 squares. How many toothpicks does Lan use altogether?

Guided solution.

Step 1. Find the pattern

Squares	1	2	3	n
Toothpicks	4	7	10	$3n + 1$

Explanation: count each kind of toothpick separately. Horizontal toothpicks: each square has 1 on top and 1 at the bottom, that is $2n$ toothpicks. Vertical toothpicks: n squares have $n + 1$ vertical toothpicks (just as n gaps need $n + 1$ posts). Total $2n + (n + 1) = 3n + 1$.

Step 2. Part a

$n = 20$: $3 \times 20 + 1 = 61$ toothpicks.

Step 3. Part b – sum of an evenly spaced sequence

The numbers of toothpicks are $4, 7, 10, \ldots, 31$ (a row of 10 squares needs $3 \times 10 + 1 = 31$ toothpicks). This is an evenly spaced sequence with step 3 and 10 terms.

Recall. Sum of an evenly spaced sequence $= \dfrac{(\text{first} + \text{last}) \times \text{number of terms}}{2}$.

Sum: $\dfrac{(4 + 31) \times 10}{2} = \dfrac{350}{2} = 175$ toothpicks.

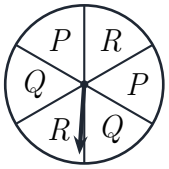
Check: $4 + 7 + 10 + 13 + 16 + 19 + 22 + 25 + 28 + 31$: pair first with last, $4 + 31 = 7 + 28 = 10 + 25 = 13 + 22 = 16 + 19 = 35$, giving 5 pairs, $5 \times 35 = 175$ ✓.

Answer: a) 61 toothpicks; b) 175 toothpicks.

Core practice

Problem 1

The spinner shown is divided into 6 sectors of equal size, each marked with a letter. The spinner is spun once. What is the probability that the arrow lands on a sector marked with the letter P?



(A) $\frac{3}{6}$ (B) $\frac{4}{6}$ (C) $\frac{5}{6}$ (D) $\frac{2}{6}$ (E) $\frac{1}{6}$

Your work:

Problem 2

Using the three digits 4, 7 and 9, each digit used *at most once*, how many three-digit whole numbers with different digits can be made?

(A) 6 (B) 3 (C) 5 (D) 12 (E) 9

Your work:

Problem 3

An envelope contains 2 sunflower seeds, 3 mung bean seeds and 4 pumpkin seeds. Chi takes one seed from the envelope at random. What is the probability that Chi takes a sunflower seed?

(A) $\frac{2}{9}$ (B) $\frac{5}{9}$ (C) $\frac{9}{7}$ (D) $\frac{7}{9}$ (E) $\frac{1}{9}$

Your work:

Problem 4

A cinema has 11 rows of seats, numbered from 1 to 11. Each odd-numbered row has 15 seats and each even-numbered row has 16 seats. How many seats are there in the cinema altogether?

(A) 176 (B) 186 (C) 165 (D) 170 (E) 171

Your work:

Problem 5

At a class party, each student in turn draws a wrapped present at random from a bag. The presents are only books and calculators; there are 27 presents in the bag altogether. Mai draws first. The probability that Mai draws a book is $\frac{2}{3}$. How many books are in the bag?

(A) 15 (B) 9 (C) 21 (D) 7 (E) 18

Your work:

Problem 6

A crate of fruit contains 20 apples, 10 oranges and no other fruit. One fruit is taken from the crate at random. What is the probability that it is an orange?

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$ (C) $\frac{1}{30}$ (D) $\frac{1}{3}$ (E) $\frac{2}{3}$

Your work:

Problem 7

A bag contains 2 red balls, 5 yellow balls and 4 blue balls, all the same size. Without looking, one ball is taken from the bag. The probability of taking a yellow ball is

(A) $\frac{2}{11}$ (B) $\frac{5}{11}$ (C) $\frac{4}{11}$ (D) $\frac{6}{11}$ (E) $\frac{7}{11}$

Your work:

Problem 8

You are doing a multiple-choice test. On one question you do not know the answer, so you choose an option at random. If the question has five options (A, B, C, D, E) and only one of them is correct, what is the probability that you guess correctly?

(A) $\frac{1}{5}$ (B) $\frac{5}{5}$ (C) $\frac{4}{5}$ (D) $\frac{2}{5}$ (E) $\frac{3}{5}$

Your work:

Problem 9

A fair six-sided die has the numbers 1 to 6 on its faces. The die is rolled once. What is the probability that it shows a 5?

(A) $\frac{2}{6}$ (B) $\frac{1}{6}$ (C) $\frac{5}{6}$ (D) $\frac{3}{6}$ (E) $\frac{4}{6}$

Your work:

Problem 10

Five friends An, Hà, Chi, Minh and Tùng sit in a row of five seats. An does *not* sit next to Hà, and Chi sits next to Minh. Who *cannot* sit in the middle seat?

(A) An (B) Hà (C) Chi (D) Minh (E) Tùng

Your work:

Problem 11

A box of jelly beans contains 15 red, 20 blue and 16 green jelly beans. Khoa takes a green jelly bean and eats it. Then he takes a blue jelly bean and eats it too. If each jelly bean left in the box is equally likely to be chosen, what is the probability that the next jelly bean Khoa takes is red?

(A) $\frac{15}{31}$ (B) $\frac{34}{49}$ (C) $\frac{15}{49}$ (D) $\frac{2}{7}$ (E) $\frac{1}{15}$

Your work:

Problem 12

Nam is counting the sheep in a flock as they cross a road. The sheep start crossing at 2:00 p.m., at a steady rate of 3 sheep per minute. After counting 42 sheep, Nam falls asleep. He sleeps for exactly an hour and a half and then wakes up, and at that moment the number of sheep that have crossed since 2:00 p.m. is exactly half of the whole flock. How many sheep are in the whole flock?

(A) 630 (B) 621 (C) 582 (D) 624 (E) 618

Your work:

Problem 13

A positive whole number is called a *palindrome* if its digits read the same from left to right as from right to left. For example, 474 and 222 are palindromes. How many palindromes are there between 100 and 1000?

(A) 10 (B) 90 (C) 100 (D) 900 (E) 1000

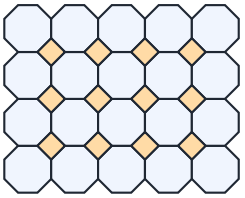
Your work:

Problem 14

The figure shown is made only of octagons and small squares. The ratio of the number of octagons to the number of squares is

(A) 1 : 1 (B) 2 : 1 (C) 25 : 12

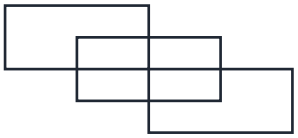
(D) 5 : 0 (E) 5 : 3



Your work:

Problem 15

The figure shown consists of three identical rectangles overlapping each other. How many rectangles of all sizes are there in the figure?



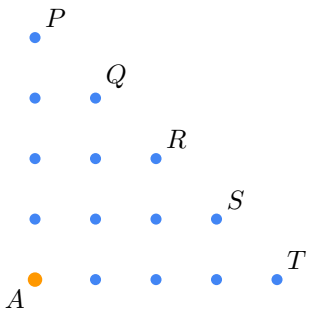
(A) 11 (B) 15 (C) 7

(D) 13 (E) 9

Your work:

Problem 16

On the dot grid shown, Lan stands at dot A. She tosses a fair coin to decide how to move: if the coin shows *heads* she moves *up* one dot; if it shows *tails* she moves *right* one dot. After exactly four tosses, Lan will stop at one of the five dots P, Q, R, S, T. What is the probability that Lan stops at dot R?



(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{9}{16}$ (D) $\frac{7}{16}$ (E) $\frac{5}{16}$

Your work:

Problem 17

Identical toothpicks are used to build rectangular grids of squares, as shown below. For example, the 1×10 grid (1 row of 10 squares) uses 31 toothpicks in total. How many toothpicks are needed for a 43×10 grid (43 rows, each with 10 squares)?



(A) 913 (B) 860 (C) 871 (D) 903 (E) 946

Your work:

Problem 18

In the cafeteria at Minh's school, the meats are beef and chicken, and the fruits are apple, pear and banana. Minh is given a random lunch consisting of one meat and one fruit. The probability that Minh's lunch includes a banana is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{5}$ (E) $\frac{3}{5}$

Your work:

Problem 19

A special six-sided die is made, with one number written on each face (numbers may repeat). When this die is rolled, the probability of getting a multiple of 3 is $\frac{1}{2}$, and the probability of getting an even number is $\frac{1}{3}$. The six numbers on the faces could be

(A) $1, 2, 3, 5, 5, 6$ (B) $1, 2, 3, 3, 5, 6$ (C) $1, 2, 3, 4, 6, 6$

(D) $1, 2, 3, 3, 4, 6$ (E) $2, 3, 3, 3, 5, 6$

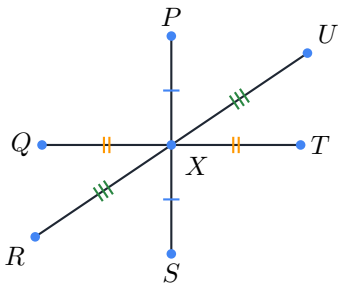
Your work:

Problem 20

Three bus routes run along the roads shown in the figure, all passing through stop X (segments with the same number of small marks are equal: $PX = XS$, $QX = XT$, $RX = XU$).

- Bus A runs the route $P \to X \to S \to X \to P$; each round trip takes 12 minutes.
- Bus B runs the route $Q \to X \to T \to X \to Q$; each round trip takes 20 minutes.
- Bus C runs the route $R \to X \to U \to X \to R$; each round trip takes 28 minutes.

At 1:00 p.m., buses A, B, C leave P, Q, R respectively. Each bus travels at a constant speed, turns around instantly at both ends of its route and runs without stopping until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., at how many moments are two or more buses at stop X at the same time?



(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Your work:

Problem 21

Each face of a cube is painted with exactly one colour. What is the smallest number of colours needed to paint the cube so that any two faces sharing an edge have different colours?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Your work:

Problem 22

How many different pairs (m, n) can be formed from the integers $1, 2, 3, \ldots, 20$ such that $m < n$ and $m + n$ is even?

(A) 55 (B) 90 (C) 140 (D) 110 (E) 50

Your work:

Problem 23

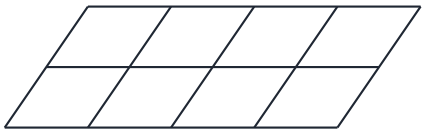
Two different two-digit positive integers are called a *reversal pair* if one is obtained from the other by swapping its two digits. For example, 52 and 25 form a reversal pair. The number 2015 has a special property: it is the product of three different prime numbers, two of which form a reversal pair ($2015 = 5 \times 13 \times 31$). Including 2015, how many positive integers less than 10 000 have this same property?

(A) 18 (B) 14 (C) 20 (D) 17 (E) 19

Your work:

Problem 24

The figure is made of 8 identical small parallelograms. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?



(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Your work:

Problem 25 *

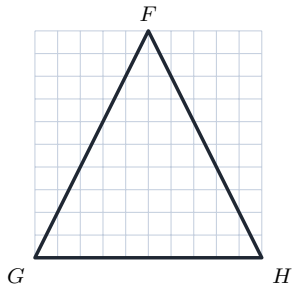
A jar contains 2 red marbles and 2 blue marbles and no other marbles. Hoa randomly takes 2 marbles out of the jar at the same time. If the two marbles have the same colour, she throws one away and puts the other back into the jar. If they have different colours, she throws away the red marble and puts the blue marble back into the jar. Hoa does this three times in total. The probability that the last marble left in the jar is red is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$ (E) 0

Your work:

Problem 26

On the 10×10 grid of squares shown, triangle FGH is drawn. Point P can be any of the 41 points where grid lines cross that lie *inside* (not on the sides of) triangle FGH. For each position of P, exactly three triangles are formed: FPG, GPH, HPF. Among all $41 \times 3 = 123$ of these triangles, how many have an area equal to exactly half the area of triangle FGH?

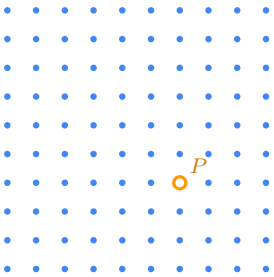


(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

Your work:

Problem 27

A 10×10 square grid is made of 100 points, as shown. Point P is marked. One of the other 99 points is chosen at random to be Q. The probability that segment PQ is horizontal or vertical is



(A) $\frac{2}{11}$ (B) $\frac{1}{5}$ (C) $\frac{1}{10}$ (D) $\frac{4}{25}$ (E) $\frac{5}{33}$

Your work:

Problem 28

The sum of all the digits of the integers from 98 to 101 is

$$9+8+9+9+1+0+0+1+0+1=38.$$

What is the sum of all the digits of the integers from 1 to 2008?

(A) 30 054 (B) 27 018 (C) 28 036 (D) 30 036 (E) 28 054

Your work:

Problem 29

Two fair dice are rolled, one red and one blue (each has six faces numbered 1 to 6). The probability that the number on the red die is greater than the number on the blue die is

(A) $\frac{18}{36}$ (B) $\frac{25}{36}$ (C) $\frac{15}{36}$ (D) $\frac{12}{36}$ (E) $\frac{17}{36}$

Your work:

Problem 30

Two fair dice are rolled, one red and one blue. The probability that the *product* of the two numbers shown is even is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$ (E) $\frac{11}{36}$

Your work:

Extension and challenge

Problem 31

A pencil case holds 6 red pens, 9 blue pens and 5 black pens, identical except for the colour of the ink. One pen is taken at random.

a) Find the probability of taking a blue pen.

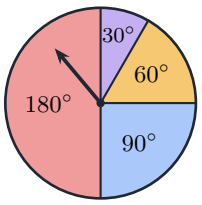
b) Find the probability of taking a pen that is *not* black.

c) How many more black pens must be put into the case so that the probability of taking a black pen is $\frac{1}{2}$?

Your work:

Problem 32

A spinner is divided into four sectors with central angles 180° (red), 90° (blue), 60° (yellow) and 30° (purple), as shown. The spinner is spun once.



a) Find the probability that the arrow lands on blue.

b) Find the probability that the arrow lands on yellow or purple.

c) Hùng says: "There are 4 colours, so each colour has probability $\frac{1}{4}$." Is Hùng right or wrong? Why?

Your work:

Problem 33

The school canteen has this lunch menu: one main dish out of 3 (chicken rice, phở, bún chả); one drink out of 2 (iced tea, orange juice); one dessert out of 2 (chè, yoghurt). Each lunch consists of one main dish, one drink and one dessert.

a) How many different lunches are there?

b) Lan is given a lunch at random. Find the probability that it includes phở.

c) Find the probability that Lan's lunch includes both phở and orange juice.

Your work:

Problem 34

A hall has 20 rows of seats. The first row has 12 seats, and each row behind has 2 seats more than the row just in front of it.

a) How many seats are in row 20?

b) Which row has exactly 30 seats?

c) How many seats does the hall have altogether?

Your work:

Problem 35

A box contains only red and blue marbles. If one marble is drawn at random, the probability that it is red is $\frac{3}{5}$. If 5 more blue marbles are put into the box, the probability of drawing a red marble drops to $\frac{1}{2}$. How many red marbles and how many blue marbles were in the box at the start?

Your work:

Problem 36

Recall: a palindrome is a number that reads the same forwards and backwards (for example 3443, 7007).

a) How many four-digit palindromes are there?

b) How many of them are even?

c) A four-digit palindrome is chosen at random. Find the probability that it is even.

d) Show that every four-digit palindrome is divisible by 11.

Your work:

Problem 37

Using the four digits $0, 1, 2, 3$, four-digit numbers with all digits different are made (each digit used exactly once).

a) How many such numbers can be made?

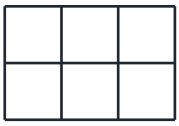
b) How many of them are even?

c) One of the numbers from part a) is chosen at random. Find the probability that it is even.

Your work:

Problem 38

The figure shows a rectangular grid of 2 rows and 3 columns of squares.



a) How many rectangles (including squares) are there in the figure altogether?

b) How many squares are there?

c) One of these rectangles is chosen at random. Find the probability that it is a square.

Your work:

Problem 39

Four friends An, Bình, Chi and Dũng are seated at random on four chairs in a row.

a) How many seating arrangements are there?

b) Find the probability that An and Bình sit next to each other.

c) Find the probability that An and Bình sit at the two ends of the row.

Your work:

Problem 40 ⁎⁎

A box contains 4 red marbles and 6 blue marbles of the same size. Hoa draws one marble at random, does *not put it back*, and then draws a second marble at random.

a) Find the probability that both marbles are red.

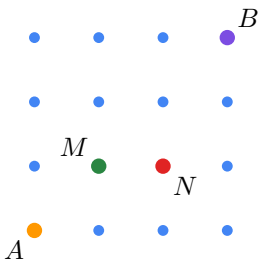
b) Find the probability that the two marbles have the same colour.

c) Find the probability that the two marbles have different colours.

Your work:

Problem 41 ⁎

On a square dot grid, Lan stands at dot A. She tosses a fair coin 6 times; each time it shows heads she moves *up* one dot, and each time it shows tails she moves *right* one dot (as in problem 16). The dots M, N, B are marked in the figure.



a) Find the probability that after 6 tosses Lan stops at dot B.

b) Find the probability that Lan passes through dot M *and* stops at B.

c) Find the probability that Lan stops at B *without* passing through dot N.

Your work:

Problem 42 ⋆⋆

We build grids of squares from toothpicks as in problem 17. An "$m \times n$" grid has m rows, each with n squares.

a) Find a formula for the number of toothpicks needed for an $m \times n$ grid.

b) A square $k \times k$ grid uses exactly 760 toothpicks. Find k.

c) Nam has exactly 2026 toothpicks and wants to build an $m \times n$ grid (with $1 \leq m \leq n$) that uses *all* of them. Find all the grids Nam can build.

Your work:

Problem 43 *

Three fair dice coloured red, blue and yellow are rolled together (each is numbered 1 to 6). Find the probability of each event:

a) the three numbers are all different;

b) the number on the red die is less than the number on the blue die, and the number on the blue die is less than the number on the yellow die;

c) at least one die shows a 6;

d) the largest of the three numbers is exactly 4.

Your work:

Problem 44 **

Special six-sided dice are made with one of the numbers $1, 2, 3, 4, 5, 6$ written on each face (repeats allowed), so that, as in problem 19, the probability of rolling a multiple of 3 is $\frac{1}{2}$ and the probability of rolling an even number is $\frac{1}{3}$. Two dice are considered the same if they carry the same six

numbers (regardless of the order of the faces).

a) What is the largest possible sum of the six numbers on such a die? What is the smallest?

b) How many different such dice are there?

Your work:

Problem 45 ★

Three lights flash: the first flashes every 4 seconds, the second every 6 seconds, the third every 10 seconds. At the start (0 seconds) all three flash together. Consider the next 10 minutes (from second 1 to the end of second 600).

a) How many times do all three lights flash together?

b) At how many moments do *exactly two* lights flash together?

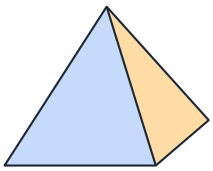
c) At how many moments does at least one light flash?

Your work:

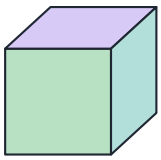
Problem 46 ★★

a) A square pyramid has 5 faces: a square base and four triangular side faces. What is the smallest number of colours needed to paint the 5 faces (one colour per face) so that any two faces sharing an edge have different colours?

b) There are 6 different colours. We paint the six faces of a cube, one colour per face, using all 6 colours. Two cubes are considered the same if one can be rotated to look exactly like the other. How many different cubes are there?



square pyramid



cube with 6 colours

Your work:

Problem 47 ★★

Consider the integers $1, 2, 3, \ldots, 20$.

a) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 3?

b) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 4?

c) How many triples (m, n, p) with $m < n < p$ are there such that $m + n + p$ is divisible by 3?

Your work:

Problem 48 *

For a two-digit number $n = \overline{ab}$, call $\overline{ba}$ the *reversal* of n (as in problem 23).

a) Find all two-digit numbers n such that n plus its reversal is a perfect square.

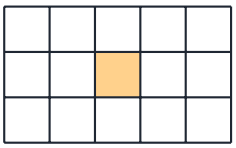
b) Find all two-digit numbers with both digits non-zero such that n minus its reversal is a positive perfect square.

c) Prove that every four-digit number of the form $\overline{abba}$ (a palindrome) is divisible by 11.

Your work:

Problem 49 **

A grid of 3 rows and 5 columns of squares (15 small squares) is shown; the square in the very centre is shaded.



a) How many rectangles (including squares) with sides on the grid lines are there in the figure?

b) How many of them are squares?

c) One of the rectangles in part a is chosen at random. Find the probability that it contains the shaded square.

Your work:

Problem 50 ★★

A box contains 2026 red marbles and 2027 blue marbles. Outside the box there is a very large pile of blue marbles. Phong repeats the following step: take 2 marbles from the box at random.

- If the two marbles have the *same colour* (both red or both blue): throw both away, then take 1 blue marble from the pile outside and put it into the box.
- If the two marbles have *different colours*: throw away the blue one and put the red one back into the box.

Phong stops when only 1 marble is left in the box.

a) How many steps did Phong carry out?

b) Find the probability that the last marble is red.

c) If the box had started with 2025 red marbles and 2027 blue marbles, what would the probability in part b be?

Your work: